Forensic Workflows for Explainability Fidelity in Domain-Specific Automated RAG Systems via FACILE

SY Mohamed
Dimitris Sacharidis
Université Libre de Bruxelles
Course INFO-H512 - Current Trends in Aritificial Intelligence

Abstract

As Retrieval-Augmented Generation (RAG) systems are increasingly deployed in high-stakes domains such as nuclear safety audits, ensuring the explainability and faithfulness of LLM outputs is critical. However, a "category error" in evaluation often prioritizes raw operational precision over causal transparency. We present an implementation of the FACILE (Factorized Context Interactions for LLM Explainability) framework (Nematov et al. 2025) to evaluate technical audit alerts, shifting the evaluation paradigm from predictive accuracy to Explainability Fidelity (LLMX). By employing Weakly Supervised Shapley (WSS) sampling and a Factorization Machine surrogate, we extract the interaction matrix (F). We demonstrate that a marginal plateau in raw accuracy is a strategic trade-off for a massive human-in-the-loop efficiency gain. Analyzing this matrix enables human auditors to rapidly distinguish between synergistic evidence, redundant information, and unverified hallucinations (Huo et al. 2026; Zhang et al. 2025), effectively providing an actionable diagnostic tool for high-stakes verification.

Code — https://github.com/symoh27/INFO-H512-LLMX_RAG.git

1. Introduction

The adoption of Retrieval-Augmented Generation (RAG) in critical industries requires rigorous verification mechanisms. In nuclear technical audits use case, an LLM-as-a-judge might correctly flag an anomaly, but human auditors must verify the exact source of this alert. Thus, limiting the evaluation to an operational precision can be insufficient for trust-sensitive domains. This project formulates a forensic workflow integrating the FACILE attribution method (Factorized Context Interactions for LLM Explainability) developped by (Nematov et al. 2025). We address the "Accuracy vs. Interpretability Fallacy" by prioritizing "Human-in-the-Loop" (HITL) audit efficiency over raw predictive accuracy. Rather than acting as a simple black-box scoring function, FACILE serves as an Explainability Layer, providing a statistically grounded, causal evidence to validate system KPIs, separating genuine multi-hop reasoning from keyword co-occurrence bias and hallucinations.

2. Problem Formulation and Approach

Our primary question is: *How can a Factorization Machine (FM) surrogate shift the evaluation of a domain-specific RAG setting from raw operational precision to Explainability Fidelity (LLMX)?* By extracting the interaction matrix (F) from the FM, we map document relationships—such as isolating numerical contradictions across disjoint nuclear reports (e.g., $9000m^3$ vs $10000m^3$ of contaminated soil)—to generate actionable diagnostic traces.

Crucially, this approach moves beyond treating attribution as absolute "proof." Instead, by contrasting the surrogate's mathematical synergies against physical ablation tests, we evaluate how heavily the LLM relies on grounded context versus its own parametric memory. Ultimately, this reframes reduced raw accuracy as a "Transparency Gain," prioritizing the reduction in "Time-to-Verification" for human auditors and providing a rigorous defense against LLM self-rationalization in safety-critical audits.

3. Exhaustive Audit and LLM-as-a-Judge Baseline

The data (pdf documents related to nuclear reports) is ingested, processed and structured, then organized into hierarchical graph-based clusters of chunks (RAPTOR like approach). Then the RAG pipeline moves into the evaluation phase. This phase relies on a dual-LLM architecture: an *Auditor* LLM for exhaustive anomaly detection, followed by an independent *Judge* LLM for forensic verification. Instead of a classical QA RAG, this audit system is performed in an automated loop, by iterating in all the critical target passages from the dataset to be evaluated by the dual-LLM architecture.

3.1 Exhaustive Multi-Anomaly Detection

The prompt is configured to prevent the common LLM pitfall of early stopping—where the model detects a single error and ignores subsequent issues within the same document passage.

- **Exhaustivity Enforcement:** The system enforces a hard constraint requiring the LLM to search for up to 3 distinct anomalies per passage. This ensures comprehensive coverage of complex technical statements.

- **Structured Logic Explanation:** To facilitate downstream parsing and forensic analysis, the system generates a structured explanation for each detected anomaly. This includes a **Global Logic** summary, followed by a detailed breakdown into **Factual**, **Semantic**, and **Causal** analysis, culminating in a precise identification of the inconsistent statement and its textual proof.

Example of an Auditor's Output:

GLOBAL LOGIC: The document states that the AFCN has issued an opinion on the second version of the National Program for the Management of Spent Fuel and Radioactive Waste, as required by the law of August 8, 1980, but the document does not provide sufficient information to meet the expectations of the national program, particularly regarding the important deadlines and clear calendars that will allow these deadlines to be respected.
FACTUAL ANALYSIS: The document states that the AFCN has issued an opinion on the second version of the National Program... However, the document does not provide sufficient information to meet the expectations of the national program, particularly regarding the important deadlines and clear calendars.
SEMANTIC ANALYSIS: The terms 'important deadlines' and 'clear calendars' are not clearly defined in the document, which could lead to confusion and inconsistency in the implementation of the national program.
CAUSAL ANALYSIS: The lack of clear deadlines and calendars could lead to delays and inefficiencies in the management of spent fuel and radioactive waste.
TARGET DOCUMENT: Avis-AFCN-Programme-National-gestion-combustibles-uses-dechets-radioactifs-2024_INJECTED.pdf
STATEMENT: insufficient information
TEXT PROOF: "...Un avis sur la deuxième version du Programme national de gestion des combustibles usés et des déchets radioactifs, tel q..."

3.2 Forensic Verification via LLM-as-a-Judge

To mitigate false positives and hallucinations generated by the Auditor, the pipeline employs a secondary *Judge* LLM. The Judge evaluates the generated alerts using a rigorous, multi-step forensic framework (mimicking Explainability Fidelity evaluation):

- **Atomic Facts Verification:** The Judge breaks down the Auditor's claim into atomic, verifiable facts, checking each against the provided document context.
- **Causal Ablation Simulation:** A critical verification step involves a "what-if" causal analysis. The Judge assesses whether the anomaly would still exist if specific sentences in the source document were altered or removed.
- **Final Verdict and Justification:** Based on the atomic checks and causal analysis, the Judge renders a final binary verdict (`VERIFIED_ANOMALY` or `FALSE_ALARM`) along with a detailed justification. This rigorous dual-LLM workflow guarantees that only mathematically and logically sound alerts proceed to the FACILE arbitrator for attribution.

Example of a Verified Audit Alert (from baseline):

[I2 - High Certainty]
JUDGE: REAL_ANOMALY — **VERIFIED:** YES — **NLI:** CORROBORATED
FACTCHECK: The alert mentions that the document states the AFCN has issued an opinion on the second version of the National Program... Upon examining the target text, the exact phrase is found. However, the alert also mentions 'insufficient information' regarding important deadlines and clear calendars, which is not directly supported by a specific quote but is rather an interpretation of the content.
ABLATION: If the exact citations were removed from the documents, the logical basis for the alert would indeed be compromised. However, the core issue of 'insufficient information' is more about the interpretation of what is provided rather than the direct quotes themselves. The removal of the cited text would remove the context in which the insufficiency is claimed.
SYNTHESIS: This alert identifies an issue of insufficient information, which is not explicitly listed among the Ground Truth anomalies. It points out a real concern regarding the lack of specific details, which could lead to inefficiencies in the implementation of the national program. Therefore, it is considered a REAL_ANOMALY.

4. Literature Review

Our methodology positions FACILE within a broader taxonomy of data valuation and source attribution, bridging the gap between sample efficiency and interaction awareness.

4.1 Marginal Attribution and Surrogate Models

The foundation of modern attribution relies on Shapley values. *Data Shapley* (Ghorbani and Zou 2019) and *Beta Shapley* (Kwon and Zou 2022) introduced principled frameworks for equitable valuation, but typically require a massive number of Monte Carlo samples ($O(n!)$), rendering them too expensive for rapid RAG auditing. Conversely, *ContextCite* (Cohen-Wang et al. 2024) employs a linear surrogate model (Lasso regression) which is incredibly sample-efficient. However, it assumes an additive, independent relationship between sources, failing entirely to detect complex synergies.

4.2 Beyond Marginal: Interaction and Diagnostics

Recent paradigms push beyond first-order attribution. *RepoShapley* (Huo et al. 2026) introduces the concept of coalition-aware context filtering, proving that text chunks are often interaction-dependent and must be evaluated as coalitions. Furthermore, frameworks like *CUE-R* (Jain and Vedam 2026) go beyond the final answer to measure "trace divergence" via evidence perturbations (remove, replace, duplicate), highly relevant for diagnosing Hallucinations and False Positives. Additionally, alternative metrics like *ARC-JSD* (Li et al. 2026) utilize Jensen-Shannon Divergence to measure distributional shift, offering a contrasting approach to surrogate training. Recent alternatives use fast distributional metrics, but trade off deeper interaction awareness (Li et al. 2026).

4.3 Mathematical Foundations

FACILE builds upon a rigorous framework developed by (Nematov et al. 2025), which formalizes RAG attribution by defining the utility of a document subset $S \subseteq D$ as the conditional log-likelihood of generating the exact target response R_{target}. Utilizing a teacher-forcing approach, this utility is strictly quantified as:

$$v(S) = \sum_{t=1}^{|R_{target}|} \log P(token_t | token_{<t}, Q, S) \tag{1}$$

Calculating exact Shapley values over this utility function requires an intractable $O(2^{|D|})$ LLM inferences. To bridge this, FACILE introduces a mathematically complete hybrid sampling strategy (`bf_kernelshap`), combining breadth-first exploration (Hamming distance 1 neighbors) with a KernelSHAP distribution for the remaining samples. Instead of relying on linear models that assume document independence, FACILE approximates the utility landscape by fitting a Factorization Machine (FM) surrogate. The surrogate output $\hat{y}$ decomposes as:

$$\hat{y}(x) = w_0 + \sum_{i=1}^{|D|} w_i x_i + \frac{1}{2} \sum_{i=1}^{|D|} \sum_{j=1}^{|D|} (V_i \cdot V_j) x_i x_j \tag{2}$$

where $x \in \{0,1\}^{|D|}$ is the binary mask of included documents. Here, w_i captures the main marginal effect (item attribution), while the latent factors V encode pairwise dependencies, mathematically defining the interaction matrix as $F = VV^T$.

5. Methodology and Implementation

The implementation focuses on integrating a FACILE framework in our pipeline using the `SHapRAG` package from the repository of (Nematov et al. 2025).

- **Baseline (LLM-as-a-Judge):** Our baseline is the LLM-as-a-judge itself, which performs textual Chain-of-Thought reasoning (Factcheck, Ablation, and Synthesis) but lacks actual causal verification.
- **FM Surrogate Configuration (FACILE):** We apply the `bf_kernelshap` sampling strategy ($m = 64$) and tune the latent rank (k) between 1 and 8 to optimize the R^2 metric on pairwise utility differences (`r2_delta`). Target utilities are transformed into logit space before fitting.
- **Attribution Extraction:** The total Shapley attribution ϕ_i for each document is precisely computed by combining its linear main effect and its halved sum of interactions:

$$\phi_i = w_i + \frac{1}{2} \sum_{j=1}^{|D|} F_{i,j} \tag{3}$$

Pairwise interactions are directly mapped to KPIs: Synergy ($F_{i,j} \gg 0$), Redundancy ($F_{i,j} \ll 0$), and Complementarity ($F_{i,j} \approx 0$).

6. Experimental Evaluation and Analysis

Our empirical evaluation validates the core premise of FACILE—interpreting LLM anomaly detection through causal interaction—while highlighting structural nuances in how synergistic contradictions are modeled.

6.1 Multi-Source Synergy Analysis

A fundamental contribution of this evaluation is the mathematical isolation of multi-source reasoning. Detecting a true inconsistency logically requires cross-referencing conflicting information points. For *Alert 15*, identifying a numerical contradiction, FACILE correctly isolated Documents 6 and 12. The interaction score was strongly positive ($F_{ij} = +0.392$), and the counterfactual Drop/Keep test proved that removing these documents caused the utility to plummet by -52.3%.

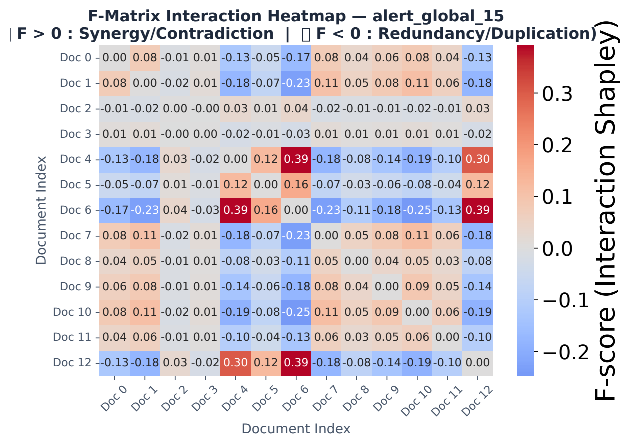

Figure 1: F-Matrix Heatmap for Alert 15, showing the strong $+0.392$ synergy between Documents 6 and 12.

The LLM actively synthesized information across two disjoint passages to deduce the anomaly. FACILE's F-Matrix cleanly captures this dependency, demonstrating its ability to discriminate alerts driven by rich, grounded reasoning from superficial keyword matching.

6.2 The Causal Disconnect (Surrogate vs. Ablation)

While the Shapley-based surrogate model captures complex interactions, physical ablation tests reveal surprising robustness in the LLM. In cases like *Alert 8* and *Alert 22*, the surrogate interaction scores are massive ($+1.660$ and $+0.818$). However, the physical Causal Impact Drop during actual ablation is negligible (-0.2% and -0.0%).

This indicates a disconnect between the surrogate explainer and the model's actual robustness. Even when mathematical synergy is present in the context, the LLM may rely on its parametric memory or semantic redundancy elsewhere in the prompt—a phenomenon known as **Contextual Entrainment**. Mechanistic studies show that LLMs can sustain responses simply because a token appeared earlier in the prompt, bypassing the need for the specific "synergistic" documents ablated. Removing the "synergistic" documents does not "break" the LLM's conclusion, challenging the assumption that interaction strictly equals causal necessity.

Table 1: Case Studies of FACILE Causal Attribution and Arbitration Decisions

Alert ID	Key Synergistic Pair	Interaction (F_{ij})	Causal Drop	Scientific Interpretation
Alert 15	Docs 6 & 12 (9000 vs 10000 m^3)	$+0.392$	-52.3%	Perfect Multi-source contradiction. Faithfully anchored and validated by ablation.
Alert 25	Docs 4 & 6	$+0.058$	-105.5%	Extreme mathematical sensitivity during ablation.
Alert 8	Docs 7 & 10	$+1.660$	-0.2%	High surrogate interaction but resilient to ablation (Parametric memory or redundancy).
Alert 22	Docs 0 & 4	$+0.818$	-0.0%	Ungrounded or highly redundant context. Ablation fails to drop utility.

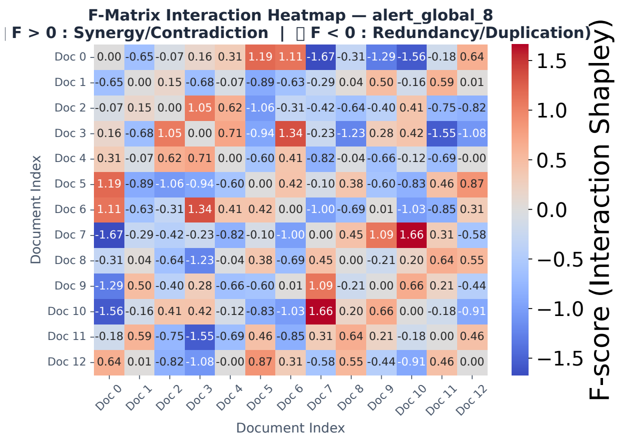

Figure 2: F-Matrix for Alert 8. Despite a massive $F_{ij} = +1.660$ between Docs 7 and 10, the actual ablation drop is -0.2%.

6.3 Methodological Vulnerabilities and Instability

While theoretically robust, the visualizations reveal specific vulnerabilities in FACILE's current implementation:

- **Numerical Variance and Mathematical Aberrations:** In *Alert 25*, the surrogate identified moderate synergy ($F_{ij} = +0.058$), but the actual ablation caused a -105.5% drop. Computing relative impacts on highly sensitive, non-linear LLM outputs can produce extreme mathematical variance. Without log-scale normalization, this instability in the utility function can lead to disproportionate penalty drops.
- **Distribution Skew and Thresholding:** The Attribution Score Distribution plot shows a heavy skew, with nearly half of the alerts clustered dangerously close to the hallucination threshold (0.1 WSS), while others stretch up to 1.6.

  A static threshold may be too rigid across diverse prompt complexities, penalizing valid alerts that simply distribute their attribution mass more widely across the context. Future implementations should apply unsupervised dynamic clustering (e.g., K-means) to the responsibility scores, an approach successfully utilized by the *RAGOrigin* framework (Zhang et al. 2025) to isolate poisoned texts.

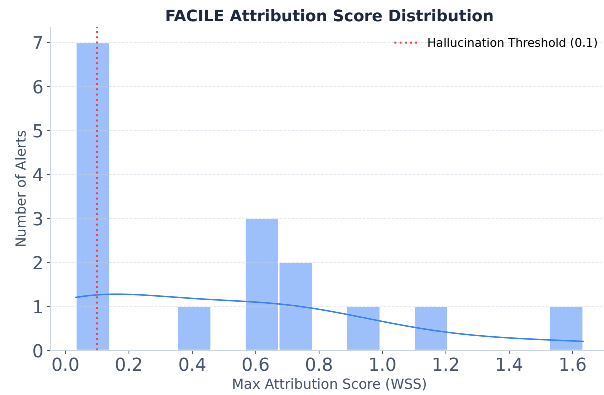

Figure 3: Global Attribution Score Distribution, highlighting the skew and thresholding challenges.

- **Directional Ambiguity of Shapley Values:** The F-Matrix heatmaps display a mix of intense red (synergy) and deep blue (redundancy). However, translating these theoretical Shapley values into human-readable trust metrics remains difficult. An auditor seeing a Drop Impact of 0.0% despite "VERY STRONG SYNERGY" faces an interpretability paradox.

7. Critical Analysis and Conclusion

While the experimental evaluation demonstrates the theoretical elegance of Factorization Machines for source attribution, evaluating our specific approach of integrating FACILE into an automated anomaly-detection RAG pipeline reveals critical insights.

Strengths and Limitations of the Integrated Approach The primary strength of our architecture lies in transforming a purely textual LLM-as-a-judge into a mathematically verifiable forensic tool. By extracting the interaction matrix (F_{ij}), we provide transparent proofs of multi-source contradictions—a necessity for highly regulated domains like nuclear compliance. However, the core limitation is computational overhead. Compared to fast heuristic or multi-agent traceability frameworks (Althaf et al. 2025), running exhaustive `bf_kernelshap` sampling and fitting a surrogate model per alert introduces significant latency, making it currently unsuitable for high-throughput, real-time alerting systems.

Key Assumptions and Their Implications Our approach relies on the fundamental assumption that surrogate interaction mathematically maps to human-interpretable logical contradictions. The implication of this assumption is exposed by the "Causal Disconnect" (Section 6.2). We assume that if F_{ij} is high, the LLM absolutely needed those documents to deduce the anomaly. However, because LLMs possess deep parametric memory and semantic redundancy, they often hallucinate or self-correct despite physical ablation. Thus, FACILE faithfully explains the *contextual prompt synergy*, but it does not fully isolate the LLM's internal cognitive failures.

Potential Improvements and Open Research Questions To transition this framework from an academic prototype to a production-ready system, several open research questions emerge. How can we implement dynamic, context-aware sampling thresholds to bypass exhaustive evaluations for low-confidence alerts? Furthermore, can this mathematical attribution be extended beyond static text documents to multi-modal, dynamic industrial data (e.g., cross-referencing real-time telemetry sensor logs with evolving safety protocols)?

Addressing these challenges presents a substantial opportunity. Scaling FACILE's mathematical transparency to diverse industrial use cases of RAG systems—specifically for automated anomaly detection in complex operational environments—represents a highly viable and impactful direction for future research.

In conclusion, our integration of FACILE into a domain-specific RAG pipeline establishes a rigorous baseline for Explainability Fidelity, proving that while computational and parametric hurdles remain, the mathematical modeling of synergistic contradictions is both feasible and necessary for trusted AI auditing.

References

Althaf, A. M.; Mohammed, M. A.; Milanova, M.; Talburt, J.; and Cakmak, M. C. 2025. Multi-Agent RAG Framework for Entity Resolution: Advancing Beyond Single-LLM Approaches with Specialized Agent Coordination. *Computers*, 14(12): 525.

Cohen-Wang, B.; Shah, H.; Georgiev, K.; and Madry, A. 2024. ContextCite: Attributing model generation to context. In *Advances in Neural Information Processing Systems*, volume 37, 95764–95807.

Ghorbani, A.; and Zou, J. 2019. Data shapley: Equitable valuation of data for machine learning. In *International Conference on Machine Learning*, 2242–2251. PMLR.

Huo, Y.; Zeng, K.; Zhang, S.; Lu, Y.; Yang, C.; Guo, Y.; and Tang, X. 2026. RepoShapley: Shapley-Enhanced Context Filtering for Repository-Level Code Completion. *arXiv preprint arXiv:2601.03378*.

Jain, S.; and Vedam, V. N. 2026. CUE-R: Beyond the Final Answer in Retrieval-Augmented Generation. *arXiv preprint arXiv:2604.05467*.

Kwon, Y.; and Zou, J. 2022. Beta shapley: a unified and noise-reduced data valuation framework for machine learning. In *International Conference on Artificial Intelligence and Statistics*, 8780–8802. PMLR.

Li, R.; Chen, C.; Hu, Y.; Gao, Y.; Wang, X.; and Yilmaz, E. 2026. Attributing Response to Context: A Jensen–Shannon Divergence Driven Mechanistic Study of Context Attribution in Retrieval-Augmented Generation. *arXiv preprint arXiv:2505.16415*.

Nematov, I.; Kalai, T.; Kuzmenko, E.; Fugagnoli, G.; Sacharidis, D.; Hose, K.; and Sagi, T. 2025. Source Attribution in Retrieval-Augmented Generation. *arXiv preprint arXiv:2507.04480*.

Zhang, B.; Xin, H.; Chen, Y.; Liu, Z.; Yi, B.; Li, T.; Nie, L.; Liu, Z.; and Fang, M. 2025. Who Taught the Lie? Responsibility Attribution for Poisoned Knowledge in Retrieval-Augmented Generation. *arXiv preprint arXiv:2509.13772*.